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 Studierichting: Informatica
 Oefeningenreeks 3G nr. 11.6

$$f(x) = \frac{x^2}{x-9}$$

$$\Rightarrow \text{VA: } x = 9$$

$$\lim_{x \rightarrow 9+} f(x) = +\infty$$

$$\lim_{x \rightarrow 9-} f(x) = -\infty$$

$$\Rightarrow \text{HA: geen}$$

$$SA : m = \lim_{x \rightarrow \infty} \frac{f(x)}{x} = \lim_{x \rightarrow \infty} \frac{x^2}{x^2 - 9x} = 1$$

$$\begin{aligned} q &= \lim_{x \rightarrow \infty} (f(x) - x) = \lim_{x \rightarrow \infty} \frac{x^2}{x-9} - x \\ &= \lim_{x \rightarrow \infty} \frac{x^2}{x-9} - \frac{x^2 - 9x}{x-9} = \lim_{x \rightarrow \infty} \frac{9x}{x-9} = 9 \end{aligned}$$

$$\Rightarrow \text{SA: } y = x + 9$$

Ligging: $\frac{x}{x \rightarrow +\infty} \left| \begin{array}{c} 9 \\ - / + \end{array} \right.$
 $x \rightarrow +\infty \Rightarrow f(x) > SA$

$$x \rightarrow -\infty \Rightarrow f(x) < SA$$

$$\begin{aligned} f'(x) &= \frac{2x(x-9) - x^2}{(x-9)^2} = \frac{2x^2 - 18x - x^2}{x^2 - 18x + 81} \\ &= \frac{x^2 - 18x}{(x-9)^2} \end{aligned}$$

$$\begin{aligned} f''(x) &= \frac{(2x-18)(x^2-18x+81) - (x^2-18x)(2x-18)}{(x-9)^4} \\ &= \frac{(2x-18)(x^2-18x+81-x^2+18x)}{(x-9)^4} \\ &= \frac{2(x-9) \cdot 81}{(x-9)^3} = \frac{162}{(x-9)^2} \end{aligned}$$

x	0			9	18		
$f(x)$	-	0	-		+	+	+
$f'(x)$	+	0	-	⁽²⁾	-	0	+
$f''(x)$	-	-	-	⁽³⁾	+	+	+
	$\nearrow$	$M(0)$	$\searrow$		$\searrow$	$m(36)$	$\searrow$
		$\cap$				$\cup$	

Grafiek:



3G-11.6-ahmed.meiyaki.INF.png

References